Pandas Dataframe

Q1. Create a DataFrame

```
import pandas as pd

data = {
    "Product": ["Laptop", "Mobile", "Tablet", "Headphones", "Keyboard"],
    "Region": ["North", "South", "East", "West", "North"],
    "Sales": [50000, 30000, 20000, 5000, 2500],
    "Quantity": [5, 10, 7, 15, 20]
}

df = pd.DataFrame(data)

print(df)
```

Q2. Data Inspection

```
import pandas as pd

df = pd.read_csv("retail_dataset.csv")

# First 5 rows
print(df.head())

# Last 5 rows
print(df.tail())

# Dataset information
print(df.info())

# Shape
print(df.shape)
```

Q3. Column Selection

```
print(df[["Product", "Region", "Sales"]])
```

Q4. Indexing using `.loc`

```
# Row with index 10
print(df.loc[10])

# Sales value at index 20
print(df.loc[20, "Sales"])
```

Q5. Indexing using `.iloc`

```
# First 6 rows and first 4 columns
print(df.iloc[:6, :4])

# Sales value from 3rd row
print(df.iloc[2]["Sales"])
```

Q6. Row Slicing

```
# First 10 rows
print(df[:10])

# Rows 15 to 25
print(df[15:26])
```

Q7. Filtering Data

```
# Region = East
print(df[df["Region"] == "East"])

# Sales > 1500
print(df[df["Sales"] > 1500])
```

Q8. Multiple Condition Filtering

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print(df[(df["Region"] == "West") & (df["Sales"] > 1200)])

Q9. Descriptive Statistics

print(df.describe())

print("Mean Sales =", df["Sales"].mean())

print("Maximum Quantity =", df["Quantity"].max())

print("Minimum Sales =", df["Sales"].min())

Q10. Aggregation Analysis

print("Total Sales =", df["Sales"].sum())

print("Average Sales =", df["Sales"].mean())

print("Maximum Sales =", df["Sales"].max())

print("Minimum Sales =", df["Sales"].min())

Expected Concepts Covered

- Creating a DataFrame
- Loading a CSV file
- `head()`, `tail()`, `info()`, `shape`
- Column selection
- `.loc` and `.iloc`
- Row slicing
- Filtering with single and multiple conditions
- `describe()`
- Aggregation methods: `sum()`, `mean()`, `max()`, `min()`